02. Conditional positivity bounds for a specified gravitational amplitude

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A low-energy theory is like a map drawn at a coarse scale: it describes what we can resolve while summarizing finer details in a few adjustable numbers. Not every choice of those numbers can come from a consistent microscopic world. This problem asks whether basic rules about probabilities, cause and effect, and scattering restrict one such number when gravity is present. Gravity's long reach makes the usual argument harder, so part of the task is making the assumptions precise.

Key terms

- **Effective field theory (EFT)** — An approximate description valid over a stated range of energies, with unresolved physics represented by additional terms.
- **Wilson coefficient** — The number multiplying one of those additional terms; it measures the strength of a correction to the simpler theory.
- **Scalar particle** — A particle with zero intrinsic spin, chosen here to simplify the scattering calculation.
- **Scattering amplitude** — A quantum quantity used to calculate probabilities for incoming particles to emerge in particular outgoing states.
- **Positivity bound** — An inequality on theory parameters derived from a quantity that must be nonnegative under specified assumptions.
- **Infrared (IR) and ultraviolet (UV)** — The low-energy or long-distance regime, and the high-energy or short-distance regime, respectively.
- **Pole and subtraction** — A pole is a singular term in a mathematical expression; a subtraction removes a specified contribution so the remaining quantity can be analyzed.
- **Dispersion relation** — A mathematical relation connecting an amplitude's behavior in different energy regions.
- **Regge assumption** — An assumption about how scattering behaves at very high energy with momentum transfer held fixed.

Why this matters

These bounds can tell us that an apparently sensible low-energy model cannot fit particular assumptions about its underlying physics. That is valuable when direct access to microscopic gravity is unavailable. A reliable bound also prevents researchers from interpreting an artifact of a singular calculation as a physical restriction.

What progress would mean

The first success would be one defensible inequality with its assumptions attached. It would narrow the parameter space of candidate theories, without claiming that every surviving theory has a complete microscopic realization.

Core question

Under a fully stated infrared prescription and high-energy assumption, what certified inequality constrains one selected low-energy coefficient of scalar scattering coupled to gravity?

Scope and assumptions

Use four-dimensional scattering of one identical real massive scalar coupled to Einstein gravity. This replaces the redundant schematic pure-gravity R^2 coefficient list with an amplitude-defined observable. Set m=1 as the mass unit; specify κ, the EFT subtraction scale, perturbative order, and whether the target is a tree coefficient or a renormalized coefficient. Graviton helicity amplitudes are a later extension.

The first task is to establish a valid sum rule. Do not run an optimization until the pole subtraction, massless cuts, crossed channel, and large-contour contribution have been derived.

Mathematical target

Let ν=s−2m^2+t/2, so s↔u sends $\nu\mapsto-\nu$. Define B(ν,t) by subtracting the explicitly calculated light poles and the chosen low-energy cut contributions from A(ν,t). Define b_2(t)=(1/2)∂^2_νB(0,t).

Derive b_2(t)=I_UV(t)+R_∞(t) with all kernels and subtractions written out. Choose either a finite-t smeared positive sum rule or a justified regulated forward limit. Nonnegativity must be established for the resulting partial-wave kernel; Im A(s,t)≥0 at arbitrary t<0 is not an admissible substitute. State a bound on R_∞ or retain it as an explicit unknown.

Required outputs and proof obligations

Return a derivation dossier, normalized coefficient definition, assumptions, and either a conditional inequality $b_2\geq$L or an explanation of the missing hypothesis. If optimizing, export a dual functional with certified kernel positivity and all truncation errors. A surviving coefficient is “not excluded by these tests,” not “UV complete.”

Validation and rejection controls

Check s↔u symmetry, dimensional consistency, and pole residues. Reproduce the nongravitational forward positivity argument in the $\kappa\to0$ limit. Use a fully specified example amplitude to test the subtraction algebra. Perturb a certificate beyond its margin and verify rejection.

Milestones and research extension

Baseline: one correct sum rule and its gravity-free limit. Strong: a numerically useful certified conditional bound with a finite positive margin. Research extension: improve the bound or relax a high-energy assumption; label any sharpness claim with both matching constructions and upper/lower evidence.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Tokuda, Aoki and Hirano, Gravitational positivity bounds — Regge assumptions and finite gravitational corrections.
- Alberte et al., Positivity Bounds and the Massless Spin-2 Pole — Why naive pole-subtracted forward positivity can fail.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$s+t+u=4m^2, \qquad \nu = s-2m^2+\frac{t}{2}, \qquad b_2(t)=\frac{1}{2}\partial_\nu^2 B(0,t).$$

$$B = A - A_{\text{light poles}} - A_{\text{chosen low cuts}}, \qquad b_2(t) = I_{\text{UV}}(t) + R_\infty(t).$$

The last equality is the sum rule to derive, with its kernel, subtractions and contour assumptions. Only a justified smearing or regulated forward prescription can turn it into a positivity statement. If that prescription proves $I_{\text{UV}} \geq 0$ and $R_\infty \geq -R_*$, then $b_2 \geq -R_*$. An uncontrolled contour term must remain unknown.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — operator basis (18–27).** In four-dimensional pure-gravity on-shell scattering, the displayed curvature-squared coefficients are not independent observables: field redefinitions and the Gauss–Bonnet combination matter. An amplitude coefficient avoids pretending that three basis-dependent numbers are separately bounded.

2. **Blocking — positivity and dispersion (44–68).** Fixed-t imaginary parts are not generally positive; unitarity acts on partial waves. The displayed dispersion formula omits essential crossed-channel/subtraction structure. The spin-2 pole makes the forward limit especially delicate, as the cited primary papers demonstrate.
3. **Major — placeholder physics (94–113).** The schematic stu/M_pl^2 expression is not a graviton helicity amplitude and lacks its pole structure. It must not seed a physical optimization.
4. **Major — interpretation (73–84).** Numerical unitarity checks do not establish all-energy consistency or a UV completion. The revised goal is a conditional necessary bound, with the Regge and infrared assumptions attached to the result.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Under a fully stated infrared prescription and high-energy assumption, what certified inequality constrains one selected low-energy coefficient of scalar scattering coupled to gravity?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: one correct sum rule and its gravity-free limit. Strong: a numerically useful certified conditional bound with a finite positive margin. Research extension: improve the bound or relax a high-energy assumption; label any sharpness claim with both matching constructions and upper/lower evidence.

Sources supporting the corrections

- Tokuda, Aoki and Hirano, Gravitational positivity bounds — Regge assumptions and finite gravitational corrections.
- Alberte et al., Positivity Bounds and the Massless Spin-2 Pole — Why naive pole-subtracted forward positivity can fail.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search,

and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and

resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.